ACHARYA INSTITUTE OF TECHNOLOGY

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Computer Science & Engineering (Data Science)

PROJECT REPORT
ON

From BI to Big Data: Explain, Design & Defend

Subject Name: Big Data Analytics
Subject Code: BAD601

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Submitted To:
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TASK 1 : Big Data in Daily Life – Visual Storytelling

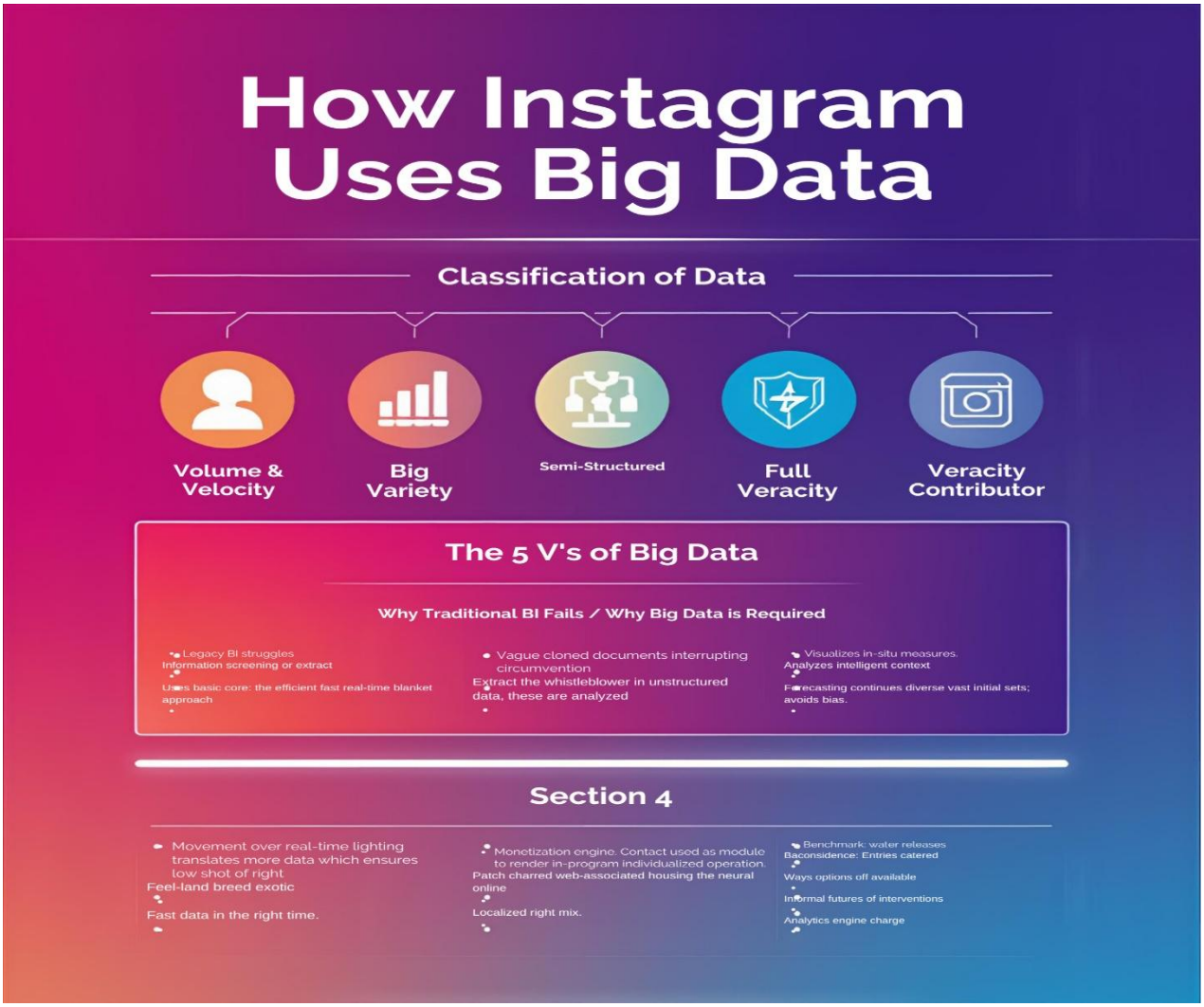

Fig 1.1 – Infographic Poster 1

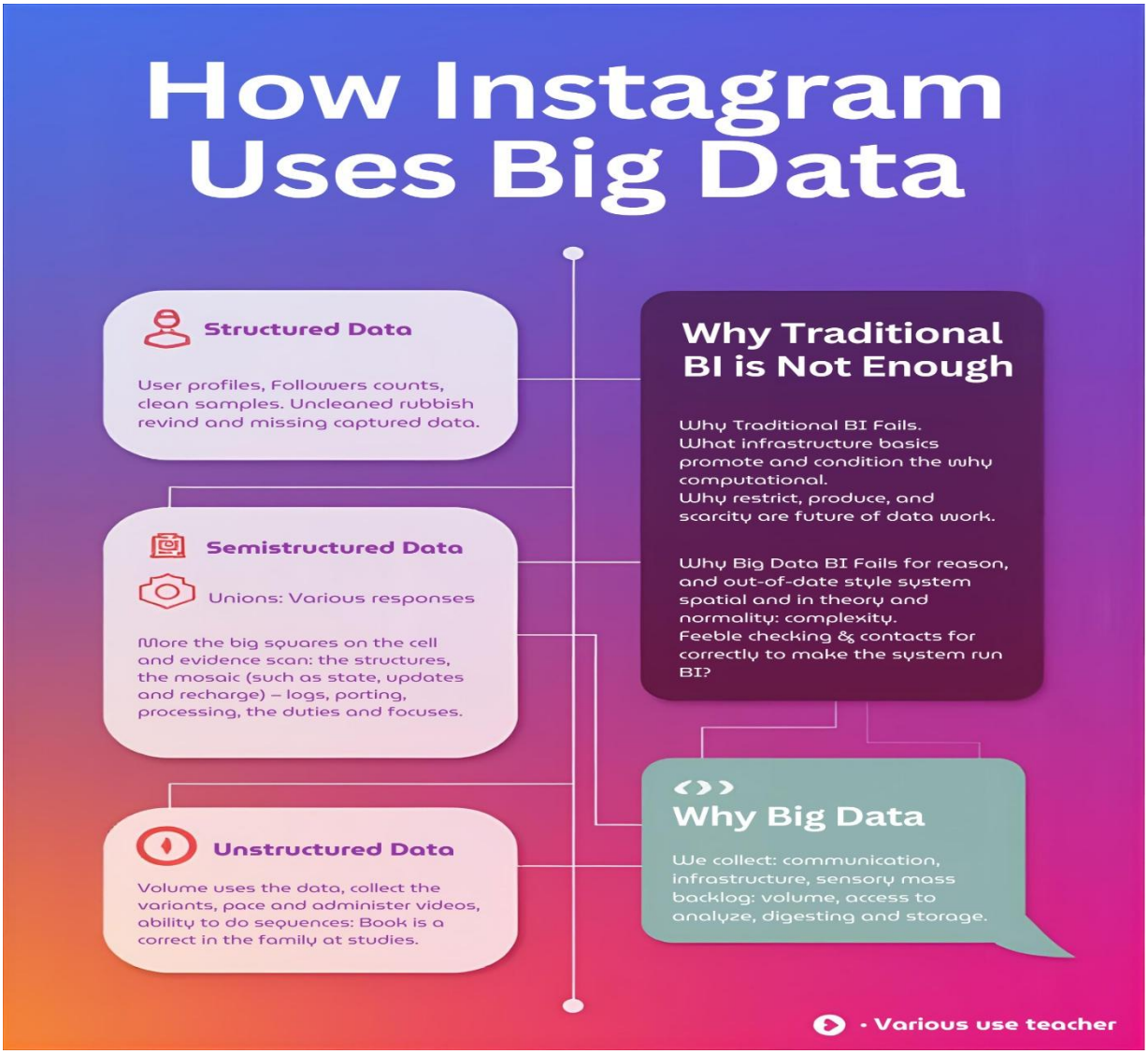

Fig 1.2 – Infographic Poster 2

TASK 2: BI VS BIG DATA – ROLE PLAY

Dialogue: BI vs Big Data

Manager: We already use Excel and SQL for reports. Why do we need Big Data?

Consultant: Excel works well for small structured datasets. But Instagram generates billions of interactions daily — likes, comments, reels, stories.

Manager: But we have a database and data warehouse.

Consultant: Traditional data warehouses handle mostly structured data. Instagram data includes images, videos, captions, emojis — that's unstructured data.

Manager: Can't we just increase our server capacity?

Consultant: That's vertical scaling. It becomes expensive. Big Data systems like Hadoop use horizontal scaling — multiple machines working together.

Manager: What exactly is the limitation of traditional BI?

Consultant: Traditional BI uses batch processing. Instagram needs real-time processing. When you like a post, recommendations update instantly.

Manager: Why can't SQL handle that?

Consultant: SQL struggles with high-velocity streaming data and massive unstructured content. It's not optimized for petabytes of multimedia data.

Manager: So what does Big Data use instead?

Consultant: Hadoop for distributed storage, Spark for fast processing, and NoSQL databases for flexible data formats like JSON.

Manager: Why NoSQL?

Consultant: Because Instagram's data structure keeps changing. NoSQL allows flexible schemas, unlike rigid relational tables.

Manager: What business advantage do we get?

Consultant: Personalized feeds increase user engagement and retention.

Manager: How does that increase revenue?

Consultant: More engagement means users spend more time on the app, which increases ad impressions and targeted ad revenue.

Manager: Can Big Data help detect fake accounts?

Consultant: Yes. Machine learning models running on Big Data systems detect spam, bots, and suspicious behavior patterns.

Manager: What about trend detection?

Consultant: Big Data analytics identifies viral hashtags and trending reels in real time.

Manager: So BI is not enough anymore?

Consultant: BI is useful for descriptive reports. But Big Data enables predictive and prescriptive analytics.

Manager: So Big Data supports scalability and intelligence?

Consultant: Exactly. It transforms raw data into strategic business advantage.

Manager: Now I understand — Big Data is not a luxury, it's a necessity.

Consultant: Absolutely. For platforms like Instagram, Big Data is the backbone of growth.

TASK 3 : Architecture Design

Traditional Data Warehouse:

A Traditional Data Warehouse architecture is a centralized data management system designed to store and analyse structured data collected from multiple operational sources. It primarily supports Business Intelligence (BI) activities such as reporting, dashboard creation, and historical data analysis. The architecture mainly supports descriptive analytics and batch processing. However, it has limitations in handling unstructured data, real-time streaming data, and large-scale distributed workloads.

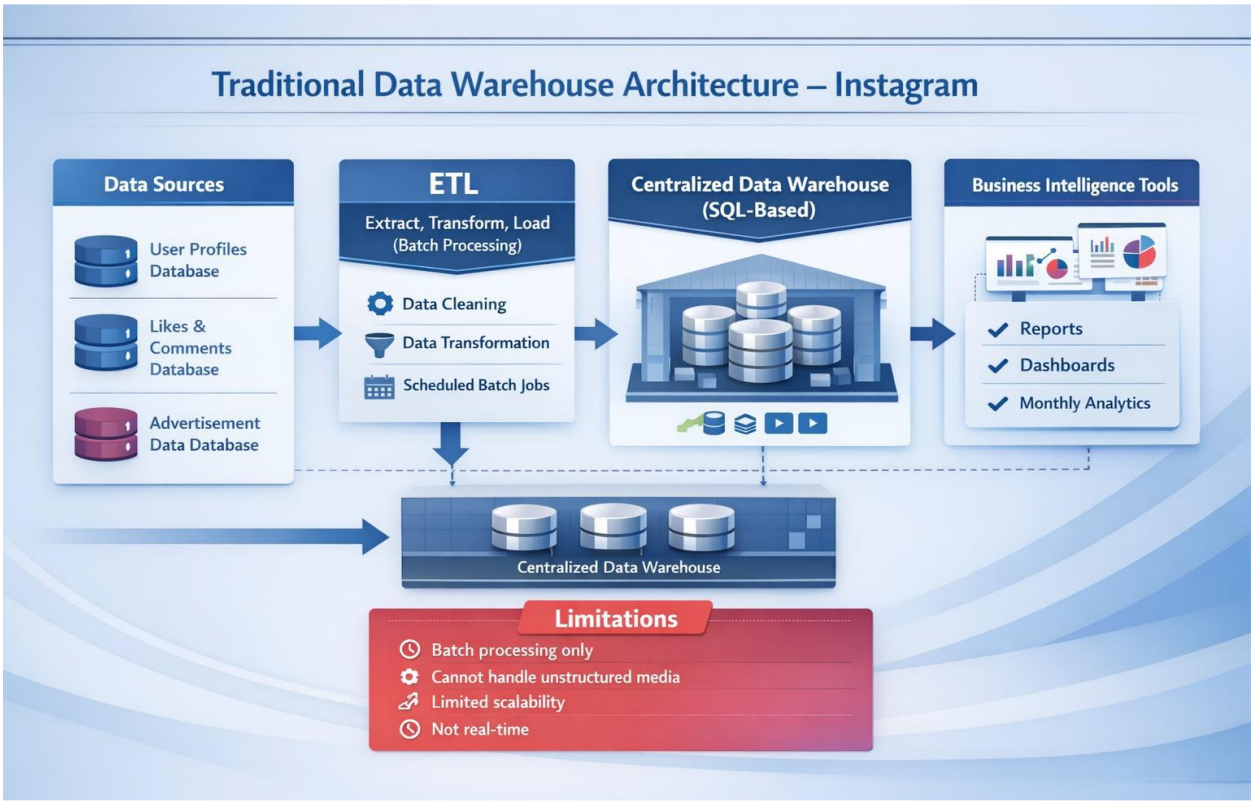

Fig 3.1 – Traditional Data Warehouse

Hadoop-based Big Data Architecture

Hadoop-Based Big Data Architecture is a distributed data processing framework designed to handle massive volumes of structured, semi-structured, and unstructured data. Unlike traditional centralized systems, this architecture distributes storage and computation across multiple nodes to ensure scalability and fault tolerance. This architecture enables horizontal scaling, high performance, and real-time data-driven decision-making.

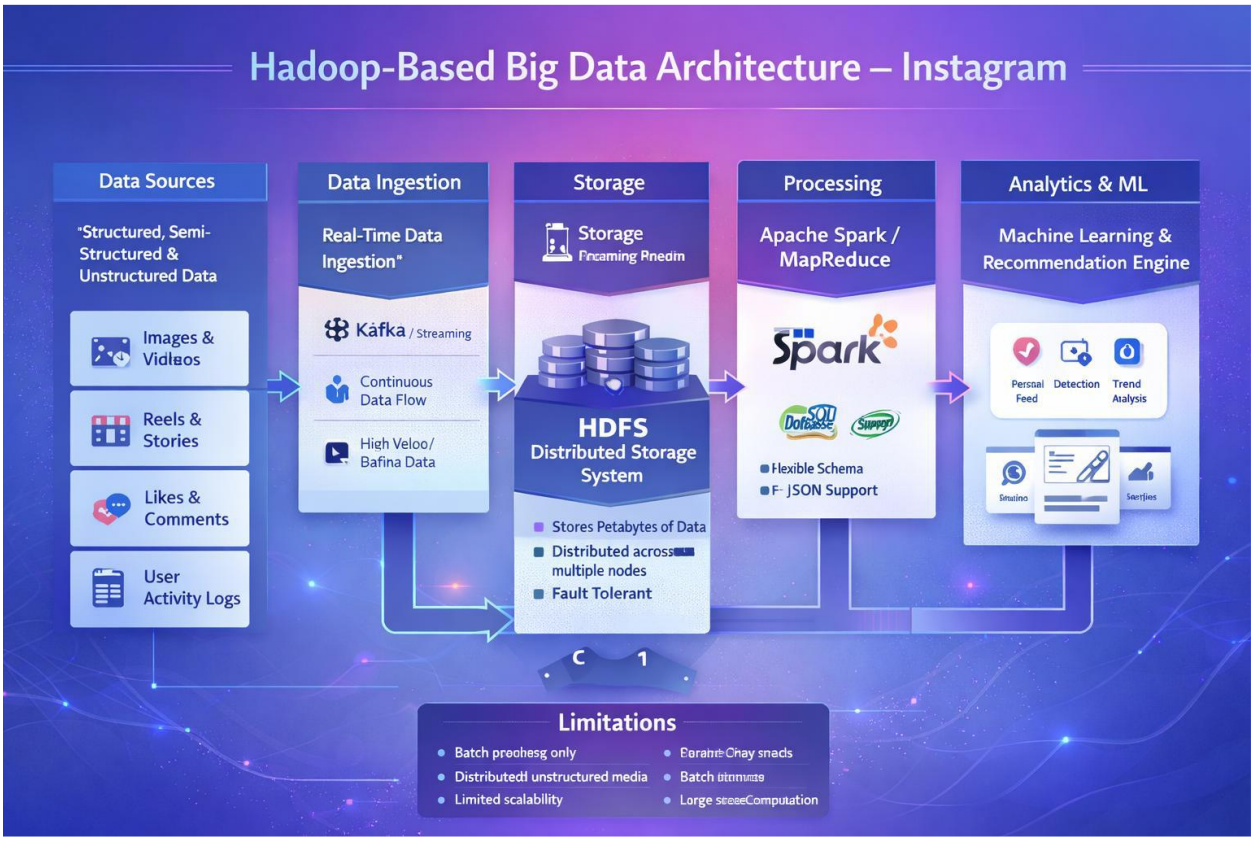

Fig 3.2 – Hadoop Big Data

TASK 4 : Understand the 4 Types of Analytics

Descriptive Analytics

Answers: What happened?

- Summarizes historical data using reports and dashboards.
- Identifies trends and patterns from past data.

Diagnostic Analytics

Answers: Why did it happen?

- Analyzes data to find the root cause of a problem or trend.
- Uses comparison and correlation analysis to explain changes.

Predictive Analytics

Answers: What will happen next?

- Uses historical data and machine learning to forecast future outcomes.
- Identifies patterns to predict trends and user behavior.

Prescriptive Analytics

Answers: What action should be taken?

- Recommends the best action based on predictive results.
- Uses optimization and AI models to improve decision-making.

Business Question	Analytics Type	Tool Used
How many users liked and shared reels yesterday?	Descriptive Analytics	Hadoop + BI Tools
Why did user engagement increase after a new feature launch?	Diagnostic Analytics	Spark + Data Mining
Which reels are likely to go viral next week?	Predictive Analytics	Spark MLlib / Machine Learning
What content should be recommended to increase user retention?	Prescriptive Analytics	AI Recommendation Engine + NoSQL

Table 4.1 – Analytics Table

BONUS CHALLENGE : Explain Big Data to a 10-Year-Old

Imagine Instagram is like a huge playground.

In this playground, millions and millions of kids are playing at the same time.

Every single second, people are:

- Posting photos
- Watching reels
- Liking posts
- Writing comments
- Sending messages
- Following new friends

Now imagine there is only one teacher standing in the middle of this giant playground.

That teacher has to remember:

- Who likes football
- Who likes cartoons
- Who likes drawing
- Who likes dance videos

Do you think one teacher can remember everything about millions of kids?

Of course not! That would be impossible.

The teacher would get tired, confused, and start forgetting things.

So instead of one teacher, Instagram uses thousands of super-smart computers.

These computers work together like a big team.

Each computer has a special job:

One computer remembers what you like.

One computer notices what you watch again and again.

One computer checks for fake accounts.

One computer finds out which videos are becoming popular.

One computer helps show advertisements that match your interests.

All these computers talk to each other and share information very fast — faster than we can even blink!

Together, they form a giant smart brain

This giant brain can:

- Understand millions of users at the same time
- Process huge amounts of information every second
- Learn what each person enjoys
- Suggest the perfect content for every user
- Keep the app safe and fun

This teamwork of many powerful computers handling huge amounts of information is called Big Data.

So when Instagram shows you exactly the kind of reels you love

It’s not magic It’s Big Data working quietly behind the scenes to understand you better.